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CONFIDENTIAL

Optotune tunable optics for Laser Processing

Enabling beam wobbling, inline inspection and 3D laser processing

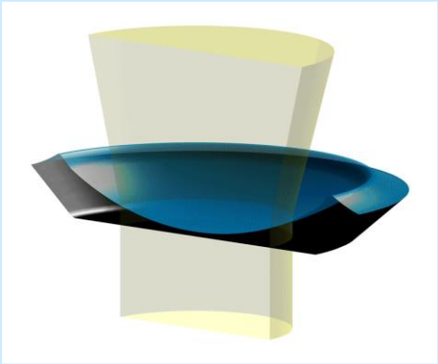
November 2024
Dr Branislav Timotijevic, BD Manager

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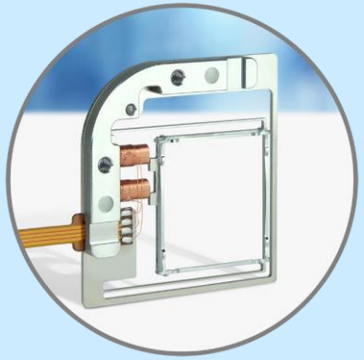
Optotune provides four core product lines

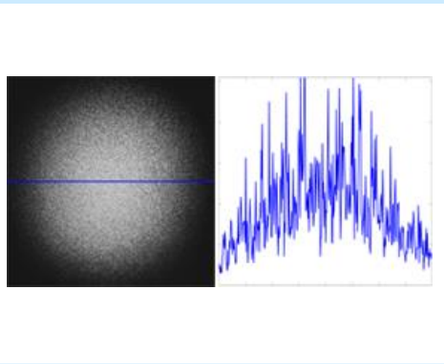
Focus tunable lenses



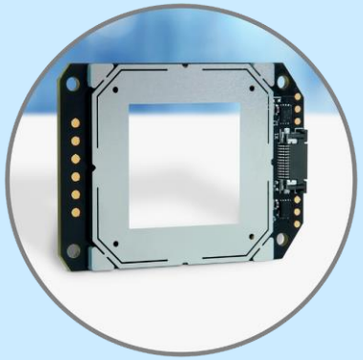


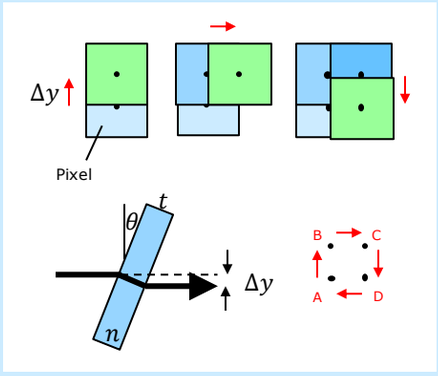
Laser speckle reducers



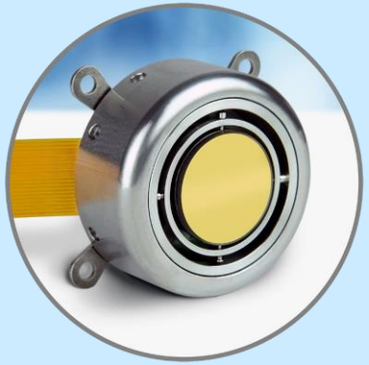


Beam shifting devices





Beam steering devices (2D mirrors)

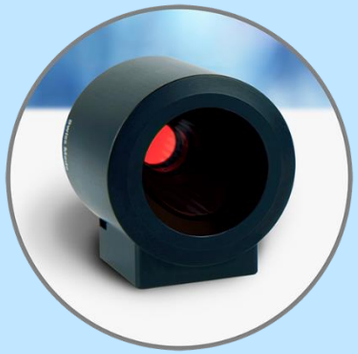


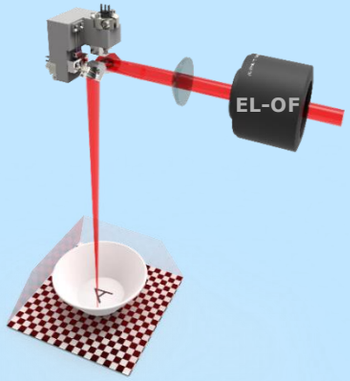


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Optotune products in Laser processing applications

2.5 & 3D laser processing



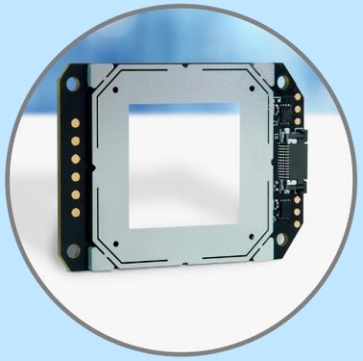


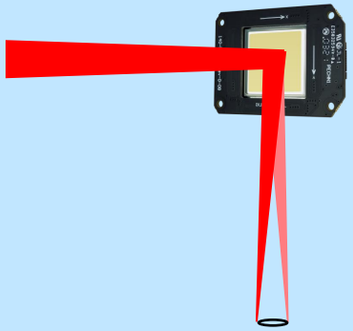
Inline inspection and AF



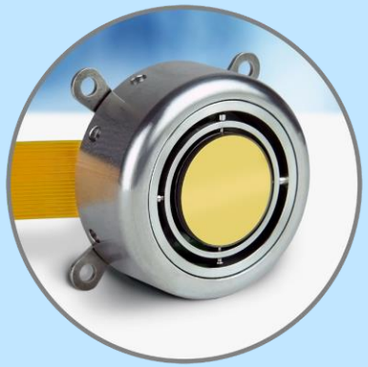


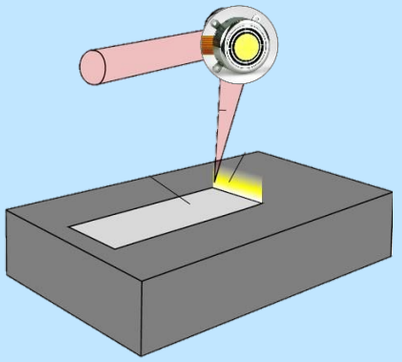
High precision 2D beam control





2D mirrors for beam steering





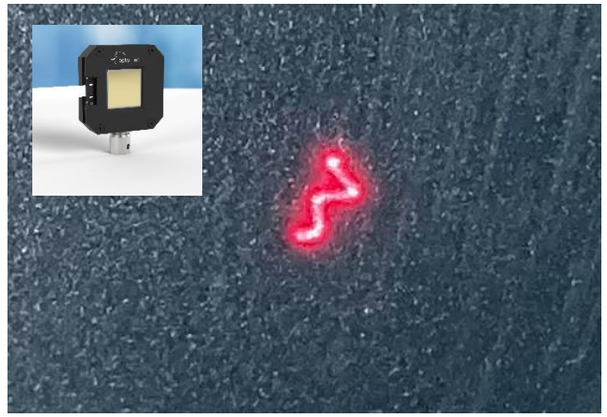
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Products, applications and benefits

FMR: Fine steering 2D mirror

- Compact laser welding
- Laser beam micro-positioning
- Laser soldering

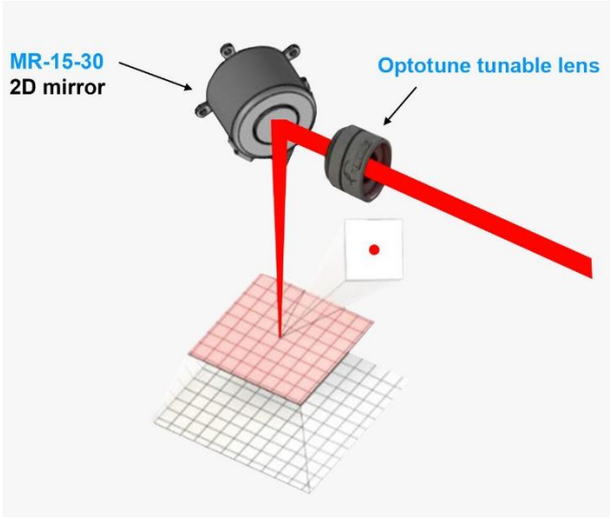


Benefits: high angular resolution, fast, 2D programmable, small, low weight, customizable, lifetime

MR: Large angle 2D mirror

- 2D beam steering at 50 W
- Free space communication
- Potential for >50 W

(tunable lens not compulsory)



Benefits: large angle, large mirror, compact package, built-in feedback, 1 optical surface for 2 DOF, lifetime

EL-16-40: Imaging lens

- Laser process inspection (both low and high power)



Benefits: compact, fast, durable, AF, distance measurement, easy to integrate in most camera systems

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Products, applications and benefits

EL-10-42: 3D laser lens

- High-end laser marking
- Laser beam focusing
- Medical lasers

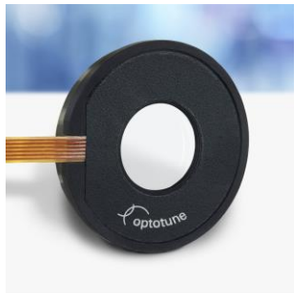


Benefits: large z-range, green and NIR, high repeatability, lifetime, compact, fast

EL-7-20 / EL-12-30 for LP

- Small dpt range laser processing < 50 W



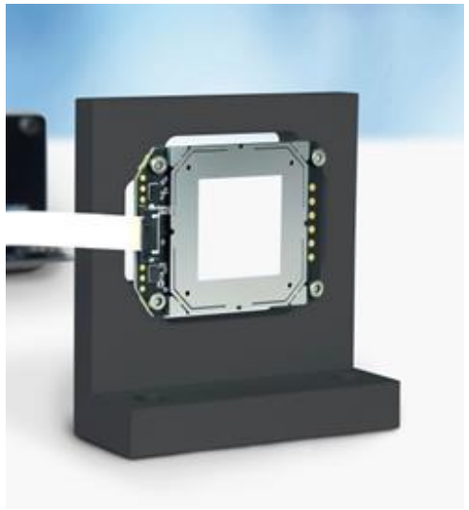




Benefits: fast, low thermal sensitivity, plano-convex to plano-concave, low power consumption, lifetime

BSW: Beam shifting window

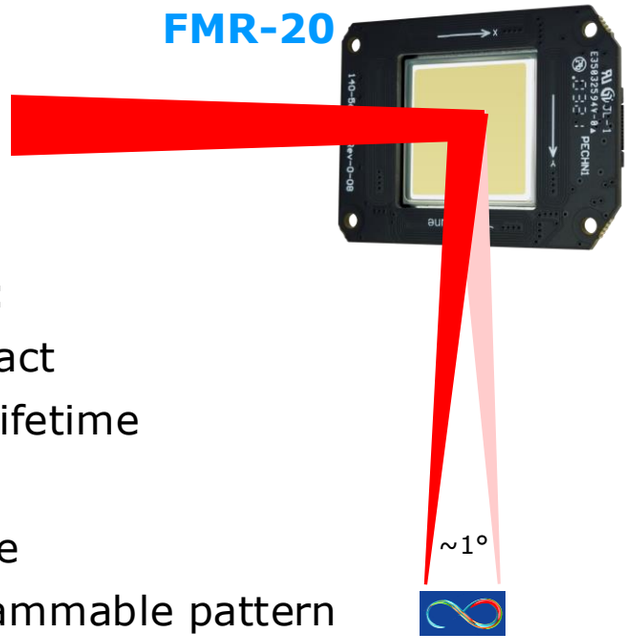
- Fibre coupling
- Colour cameras
- Hyperspectral imaging



Benefits: 2-axis beam shifting, precise, fast, high transmission, lifetime

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High-power fine beam steering with FMR 2D mirrors



Benefits:

- Compact
- Long lifetime
- Fast
- Precise
- Programmable pattern



Applications:

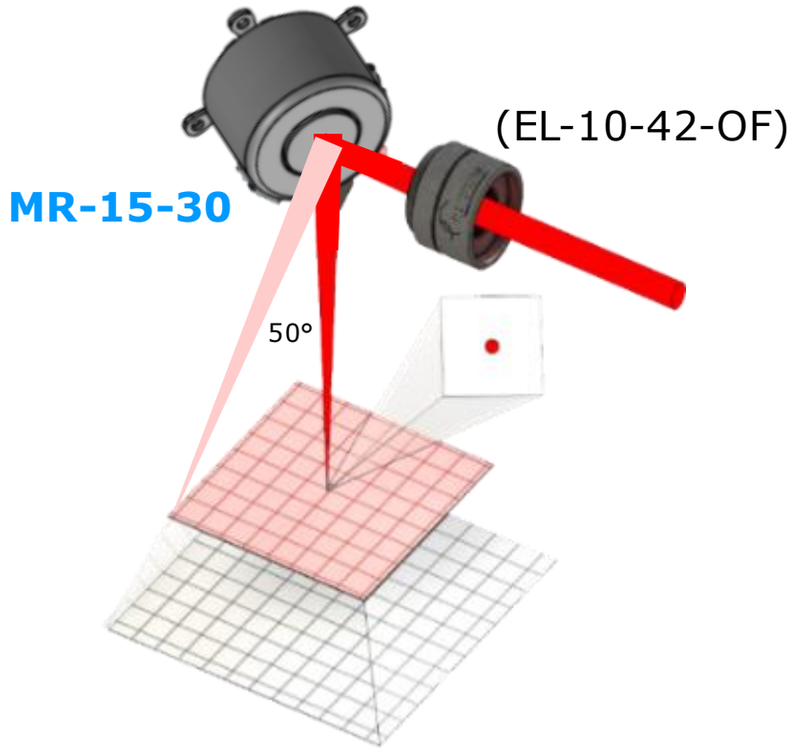
- Laser welding & soldering
- Laser cavity alignment
- Point and shoot / raster (lissajous) scanning

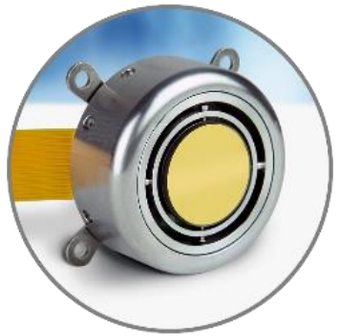
FMR-20	
Mirror size	20 x 20 mm (customizable)
Motion pattern	-, o, ∞, □ (2D programmable)
Mechanical tilt angle Single axis Both axis	 ±0.3° (5.2 mrad) ±0.2° (3.5 mrad)
Resolution (with ICC-4C-2000)	4 µrad
Bandwidth (sine wave, ± 2.3 mrad)	250 Hz
Reflectivity Protected gold (800-2000 nm,0-45°) Dielectric NIR (750-1100 nm, 0-45°) 1064 HP (1060-1100 nm, 58-70°)	 >95% >98% >99.9%
Average laser power	2 kW*
LIDT ISO 21254 (1064 nm, 10 ns, 10 Hz)	10 J/cm^2
Scale drift	1000 ppm/K
Position feedback	Open loop
Transition (rise) time	1.3 ms
Settling time	4 ms
Power consumption	< 4W
Size (without bracket)	47 x 35 x 3.65 mm
Weight (without bracket)	9 g

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Large-angle beam steering with MR 2D mirrors





MR-15-30 (quasi-static)	
Mirror size	15 mm
Mechanical tilt – fast axis (half angle)	25°
Full-scale bandwidth – fast axis	20 Hz
Mechanical tilt – slow axis (half angle)	25°
Full-scale bandwidth – slow axis	20 Hz
Mech. Repeatability RMS typical	30-100 μrad
Footprint	30x14.5
Position feedback	yes
Laser power	50 W*

Benefits:

- Large angle
- Small footprint
- Single optical surface

Applications:

- Laser templating
- Free space communication
- Low-power beam steering

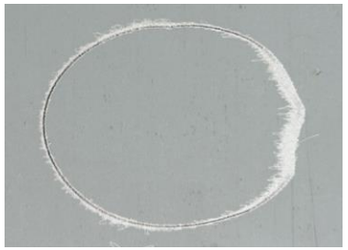
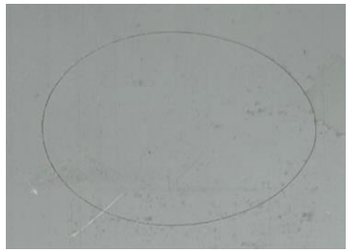
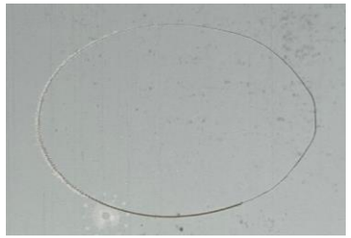
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MR-15-30-Rev2 performance up to 50 W

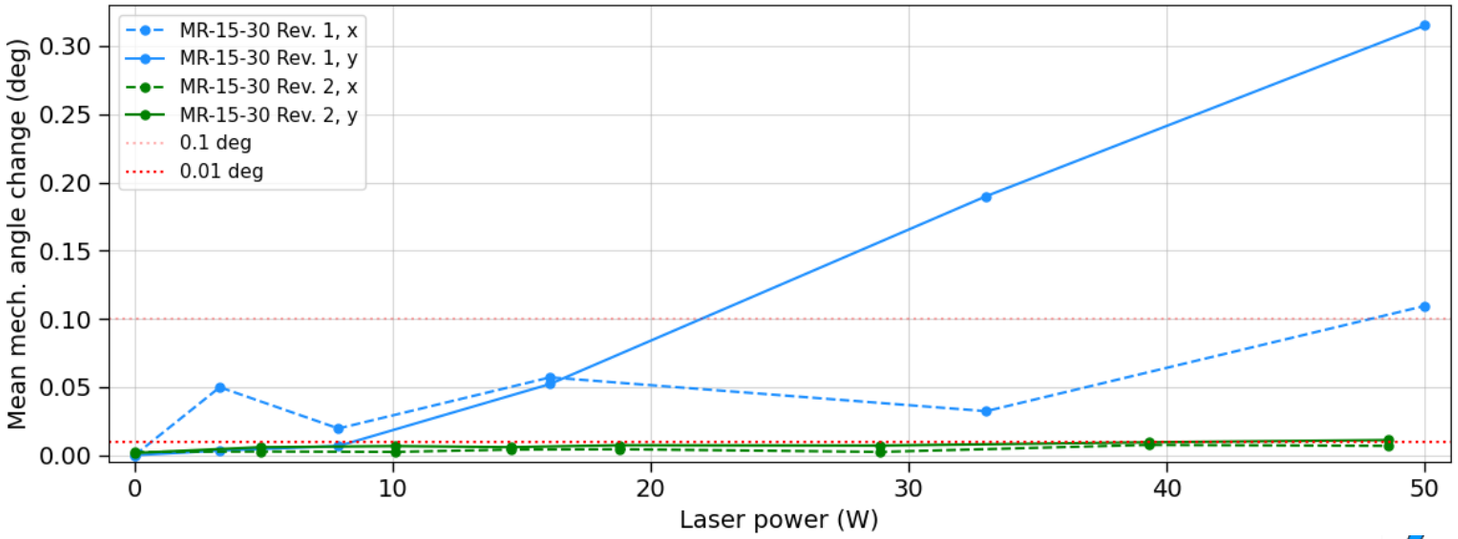
Test setup:

- Focus on metallic surface to test for the influence backscattered light on the optical feedback of the mirror
- Laser: 1064 nm, 0-50 W
- Angle of incidence: 45°
- Mirror angle (mechanical): ±5°
- Focusing lens: f=160 mm (f-theta)
- Mirror control: MR-E-2

Fast steering mirror	MR-15-30 Rev. 1	**MR-15-30 Rev. 2**
Laser 15 W Closed loop		
Laser 50 W Closed loop		

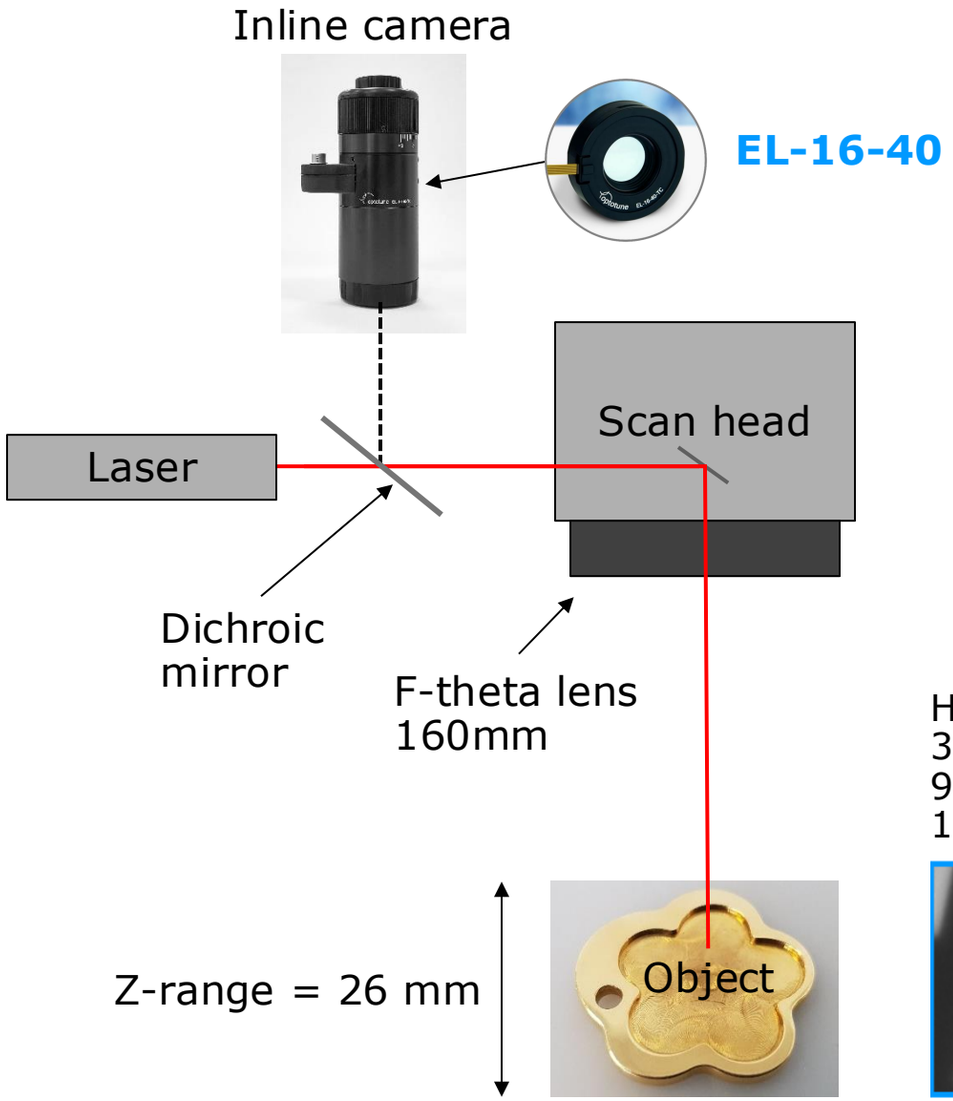
Mean absolute mechanical angle change is below <u>0.02°</u> up to 50 W

Further tests planned >50 W

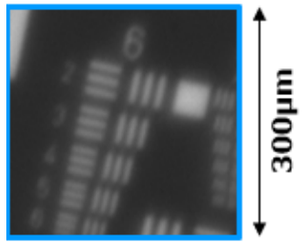


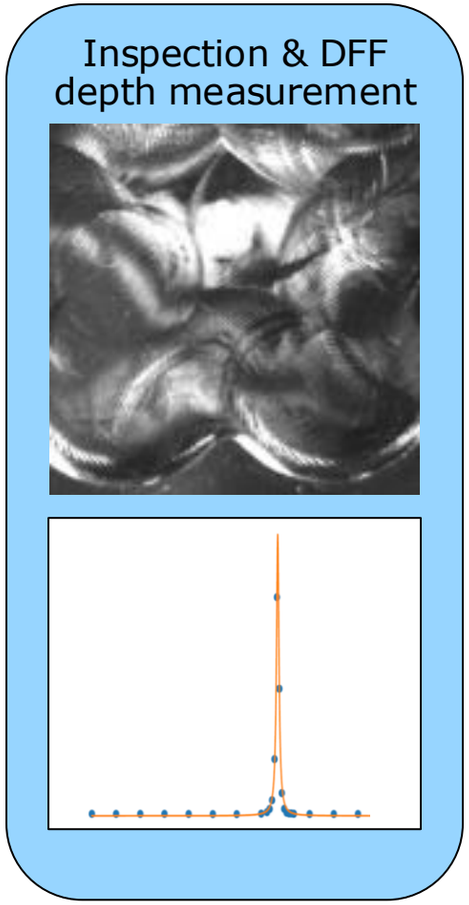
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Inline inspection with Distance measurement using EL-16-40



High resolution:
30 lp/mm (image)
9 µm on object
10 mm scan head aperture





Depth from focus: A focus tunable lens in conjunction with an autofocus algorithm can reliably measure distance to an arbitrary object in less than a second

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Inline inspection up to 50W using EL-10-42-OF



Inline camera
(fixed optics)

Laser

Scan head

Dichroic mirror

No f-theta lens

3

2

1

10 cm

Marking field

10 cm

Uniform resolution of 30 lp/mm (image) across marking field

EL-10-42-OF

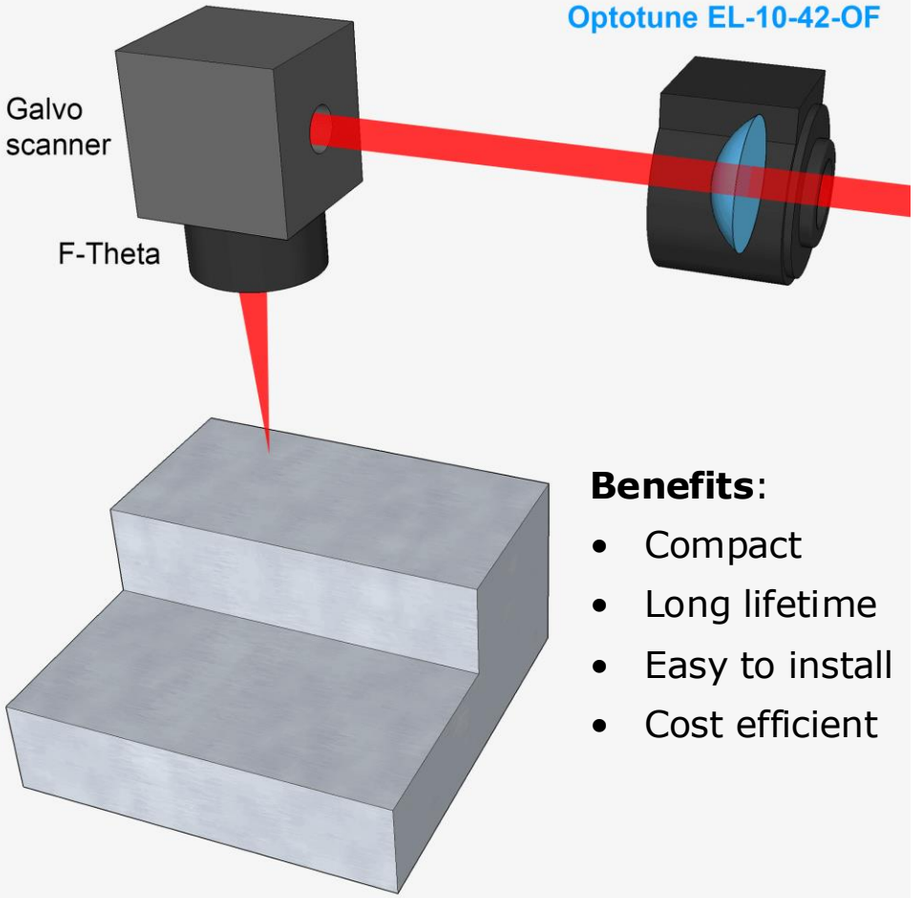
One lens, two functions:

- Laser marking
- Camera inspection

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2.5D and 3D laser processing with EL-10-42-OF

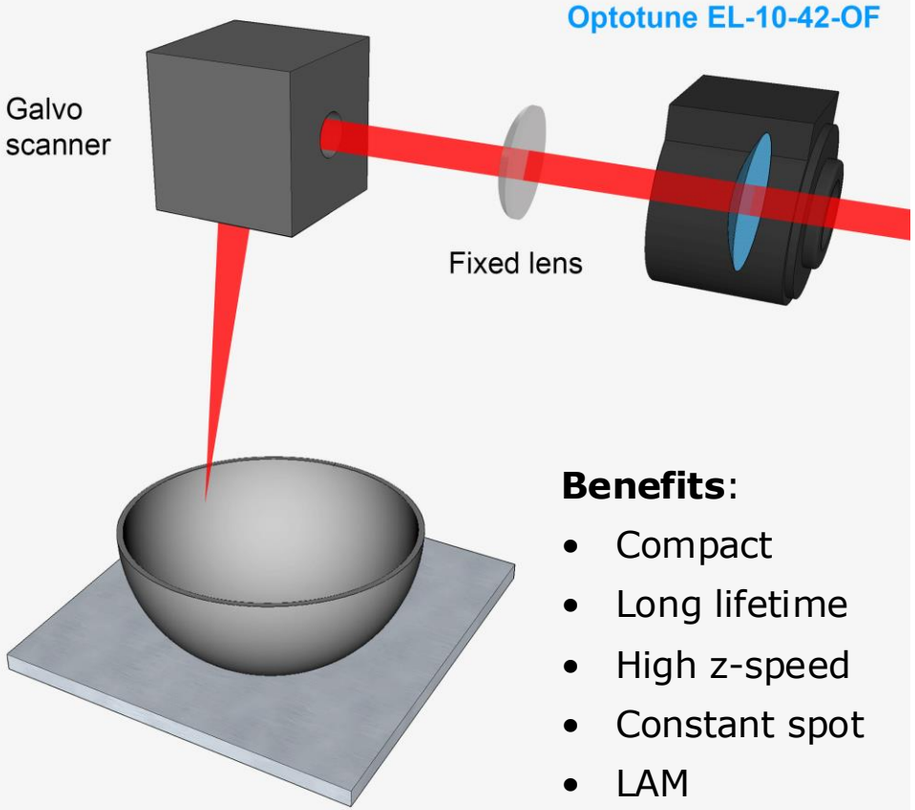
2.5D Z-stepping



Benefits:

- Compact
- Long lifetime
- Easy to install
- Cost efficient

3D laser focus control



Benefits:

- Compact
- Long lifetime
- High z-speed
- Constant spot
- LAM

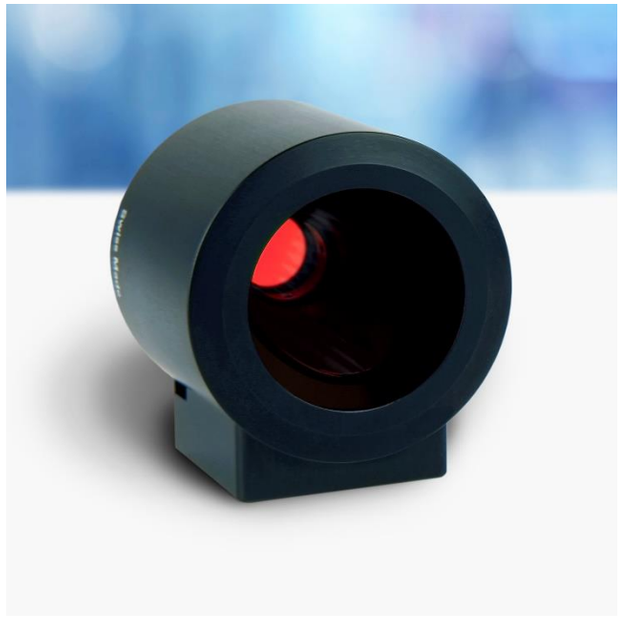
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EL-10-42-OF specs in the NIR and at 532 nm

Product	**EL-10-42-OF-NIR** **EL-10-42-OF-532**	unit
Clear aperture	10	mm
Maximum operating average laser power @ NIR (950-1100 nm) @ 532 nm	 50 20	W
Optical power: tuning range	-2.0 to +2.0	dpt
Optical power: repeatability	typical: < 0.02 max: < 0.04	dpt
Optical power: long term stability 8h		
Wavelength range (NIR)	950 – 1100	nm
Wavefront error @ 1064 nm @ 532 nm	 < 0.15 < 0.3	λ RMS
Transmission NIR (950-1100 nm) @ 532 nm	 > 94 % > 95 %	
Long term radiation damage @ 1064 nm: 40 mJ /cm2 at 20 kHz	No effect after 2000 h	
Damage threshold @ 1064 nm: 125 ns-pulsed at 50 kHz 10 ps-pulsed at 50 kHz	 2.6 2.05	J/cm^2
Response time with EL-E-OF-A analog board	80% step: 12 20% step: 6	ms
Response time with Scaps digital board	80% step: 8 20% step: 4.5	ms
Focal length resolution	Continuous (depends on control electronics)	
Lifecycles (10%-90% sinusoidal)	> 1‘000‘000‘000	

Typical parameters in a marking system with f = 160 mm f-theta lens

Max z-tuning range	100	mm
Repeatability (10%-90% step)*	typical: < 500 max: < 1000	µm
Long term drift over 8h*		



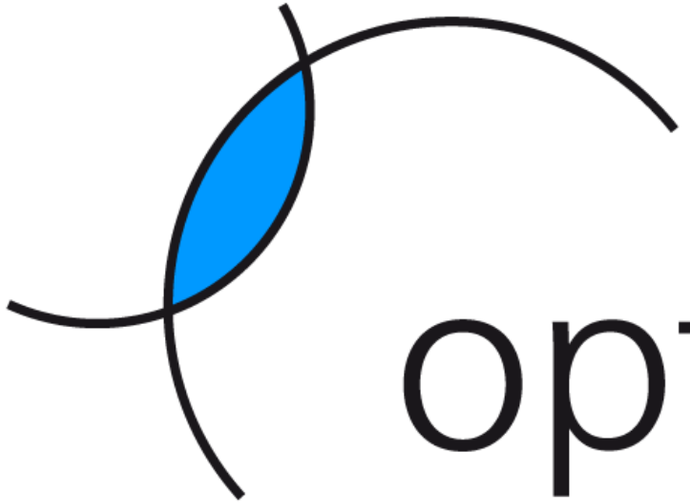
All EL-10-42-OF lenses undergo extensive OQC tests including laser testing

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Analog and digital drivers for EL-10-42-OF

	EL-E-OF-A (2.5D)	SCAPS Optotune-DSD-2-O (3D)
		
		XY2-100 integration by USB calibration interface Only one power supply Thermal Control and lens status signal
Interface	Analog 0-5V	Digital XY2-100, X-Y bi-directional Scaps interface
Controller	Microprocessor based	FPGA based
Intelligence	Standard PID control	Model based drive algorithm
80% step response	12ms	8ms
Demonstrated processing speed on 45deg slope (160mm F-Theta)	0.7m/s	6m/s
Suitable operation	Z-Stepping for 2D processing	True 3D processing

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shaping the future of optics